

ME381 ROBOTICS

Experiment No.: 5

Experiment Title: Actuators: Stepper and Servo motors

Group No.: B6

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Results for the group task:

Section 1. Stepper motor

Types ovf stepping	Number of steps required for 90-degree rotation
Full	26
Half	51
Quarter	100
One-eighth	200

Section 2. Servo with microcontroller

Sr no.	Topic	Task	Report
1.	Control servo with prebuild libraries.	Modify the code so that it creates a sweeping motion, moving the servo head from one position to another in a continuous loop.	1 (a) Modification made in the code to create the sweeping motion: Under void loop() section, “if (temp%2 == 0){ pos = 130; } else pos = 0; temp++; ”. 1 (b) when I set the value of pos to 130, the servo head rotated and pointed to 130 degrees.

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2.	Control servo with potentiometer.	Modify the code so that when the potentiometer is rotated clockwise, the servo rotates counterclockwise.	2 (a). When the potentiometer is rotated in clockwise direction, the servo also rotates in clockwise direction, and also on full rotation of the potentiometer, the servo rotates from 0 to 120 degrees because the voltage provided to the potentiometer was 3.3V only. 2 (b). Under the void loop section, I updated “int angle= map(val,0,1023,0,180);” to “int angle= map(val,0,1023,180,0);”. Basically, I reversed the mapping.
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2.

1(a).

[Video of the sweeping motion](#)

Task1_SERVO.ino

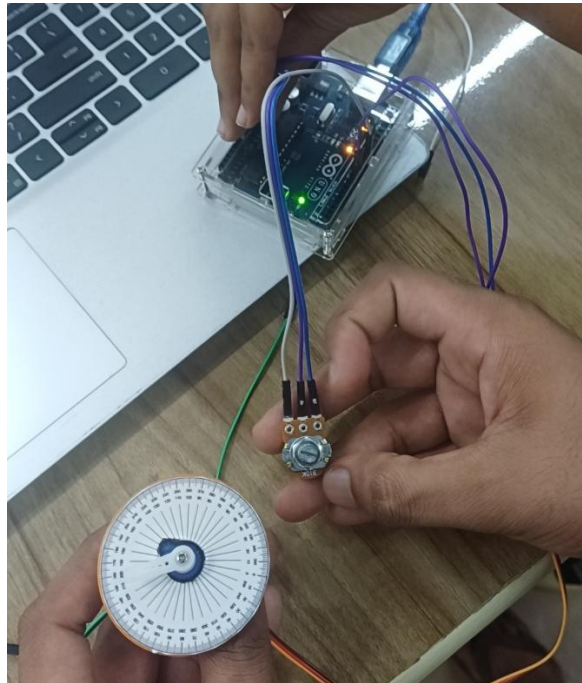
```

1  #include <Servo.h>
2
3  Servo myservo; // create Servo object to control a servo
4  // twelve Servo objects can be created on most boards
5
6  int pos = 0;    // variable to store the servo position
7  int temp = 0;
8
9  void setup() {
10     myservo.attach(9); // attaches the servo on pin 9 to the Servo object
11 }
12
13 void loop() {
14     if (temp%2 == 0){
15         pos = 130;
16     }
17     else pos = 0;
18     temp++;
19     myservo.write(pos); // tell servo to go to position in variable pos
20     delay(200);
21 }

```

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1(b).



```
Task2_SERVO.ino
1
2  #include <Servo.h>
3
4  Servo myservo; // create Servo object to control a servo
5  // twelve Servo objects can be created on most boards
6
7  int pos = 0;    // variable to store the servo position
8  int potPin=A0;  // pin attached to potentiometer
9
10 void setup() {
11     myservo.attach(9); // attaches the servo on pin 9 to the Servo object
12     pinMode(potPin,INPUT); // declare pin as INPUT
13 }
14
15 void loop() {
16     int val=analogRead(potPin); // read the analog pin
17     int angle= map(val,0,1023,180,0); // map the value of analog input to angle
18
19     myservo.write(angle);
20     delay(15); // waits 15ms for the servo to reach the position
21
22
23
Output Serial Monitor X
```